

DIGITAL SAT MATH — PRACTICE QUESTIONS

DOCUMENT 1 | QUESTIONS ONLY

Topics: Function Notation & Evaluation | Creating Linear Models from Word Problems

Total Questions: 50 MCQs | Difficulty Mix: Easy · Medium · Hard | Digital SAT Style

SECTION A — FUNCTION NOTATION & EVALUATION (Q1–Q25)

Q1. DIFFICULTY: EASY

The function h is defined by $h(x) = 7x + 3$. What is the value of $h(5)$?

- A) 35
- B) 38
- C) 43
- D) 40

Q2. DIFFICULTY: EASY

If $f(x) = 4x^2 - 2x + 1$, what is the value of $f(0)$?

- A) 0
- B) 1
- C) -1
- D) 3

Q3. DIFFICULTY: EASY

The function g is defined by $g(t) = 12 - 3t$. What is the value of $g(4)$?

- A) 24
- B) 0
- C) 6
- D) -2

Q4. DIFFICULTY: EASY

If $p(x) = 5x - 9$, for what value of x does $p(x) = 16$?

- A) 3
- B) 5
- C) 7
- D) 4

Q5. DIFFICULTY: MEDIUM

The function f is defined by $f(x) = 3x^2 + kx - 4$. If $f(2) = 14$, what is the value of k ?

- A) 0
 - B) 2
 - C) 3
 - D) 5
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Q6. DIFFICULTY: MEDIUM

If $g(x) = 2x - 5$ and $h(x) = x^2 + 1$, what is the value of $g(h(3))$?

- A) 15
 - B) 5
 - C) 17
 - D) 11
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Q7. DIFFICULTY: MEDIUM

The function f is defined by $f(x) = (x + 3)^2 - 7$. What is the value of $f(-1)$?

- A) -3
 - B) -7
 - C) 0
 - D) 4
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Q8. DIFFICULTY: MEDIUM

If $f(x) = 8 - 2x$ and $g(x) = 3x + 1$, for what value of x is $f(x) = g(x)$?

- A) $x = 1$
 - B) $x = 2$
 - C) $x = -1$
 - D) $x = 7/5$
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Q9. DIFFICULTY: MEDIUM

The function h is defined by $h(n) = n^2 - 4n + 4$. Which of the following is equivalent to $h(n)$?

- A) $(n - 2)^2$
 - B) $(n + 2)^2$
 - C) $(n - 4)^2$
 - D) $n(n - 4)$
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Q10. DIFFICULTY: MEDIUM

If $f(3) = 17$ and $f(x) = ax + b$ where a and b are constants, and $f(0) = 2$, what is the value of a ?

- A) 3
 - B) 5
 - C) 7
 - D) 15
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Q11. DIFFICULTY: HARD

The function f is defined for all real numbers by $f(x) = 2x + 3$. The function g is defined by $g(x) = f(x + 4) - f(x)$. What is the value of $g(7)$?

- A) 4
 - B) 8
 - C) 11
 - D) 3
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Q12. DIFFICULTY: HARD

If $f(x) = x^2 - 3$ and $g(x) = 2f(x) - f(2x)$, what is $g(3)$?

- A) -21
 - B) -6
 - C) 6
 - D) 18
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Q13. DIFFICULTY: HARD

The function f satisfies $f(2x) = 4f(x)$ for all values of x . If $f(1) = 5$, what is the value of $f(4)$?

- A) 20
 - B) 40
 - C) 80
 - D) 160
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Q14. DIFFICULTY: HARD

The function p is defined by $p(x) = (x - a)(x - b)$, where a and b are positive constants. If $p(0) = 12$ and $p(1) = 0$, which of the following could be true?

- A) $a = 3, b = 4$
 - B) $a = 1, b = 12$
 - C) $a = 2, b = 6$
 - D) $a = 1, b = -12$
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Q15. DIFFICULTY: HARD

The function f is defined by $f(x) = cx^2 - 5$, where c is a constant. If $f(3) = f(-3) + 12$, what is the value of c ?

- A) 1
 - B) 2
 - C) 3
 - D) No unique solution exists
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Q16. DIFFICULTY: MEDIUM

Selected values of the function f are shown in the table below.

x	$f(x)$
1	7
3	13
5	19
7	25

If f is a linear function, what is the value of $f(10)$?

- A) 31
 - B) 34
 - C) 37
 - D) 40
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Q17. DIFFICULTY: MEDIUM

If $f(x) = 2x + 1$ and $f(g(x)) = 2x + 9$, what is $g(x)$?

- A) $g(x) = x + 4$
 - B) $g(x) = x + 8$
 - C) $g(x) = 2x + 4$
 - D) $g(x) = 4x + 4$
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Q18. DIFFICULTY: MEDIUM

The function d is defined by $d(r) = \pi r^2$. If $d(r + 1) - d(r) = 7\pi$, what is the value of r ?

- A) 2
 - B) 3
 - C) 4
 - D) 5
-

Q19. DIFFICULTY: EASY

The function q is defined by $q(x) = -x + 8$. What is $q(-3)$?

- A) 5
 - B) 11
 - C) -5
 - D) -11
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Q20. DIFFICULTY: EASY

If $f(x) = 6x - 7$, what is the value of x for which $f(x) = 5$?

- A) 1
 - B) 2
 - C) 3
 - D) 4
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Q21. DIFFICULTY: MEDIUM

The function w is defined by $w(x) = |2x - 6|$. What is the smallest value of $w(x)$?

- A) 0
 - B) 3
 - C) 6
 - D) -3
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Q22. DIFFICULTY: MEDIUM

If $k(x) = 3x + 2$ and $k(m) = k(2m - 1)$, what is the value of m ?

- A) -3
 - B) 1
 - C) 3
 - D) -1
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Q23. DIFFICULTY: HARD

The functions f and g are defined by $f(x) = 3x - 1$ and $g(x) = x + 5$. For what value of x is $f(g(x)) = g(f(x))$?

- A) All real values of x
 - B) $x = 1$
 - C) $x = 6$
 - D) No solution exists
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Q24. DIFFICULTY: HARD

The function f is defined by $f(x) = x^2 + bx + c$. If $f(2) = 0$ and $f(5) = 0$, what is the value of $f(0)$?

- A) 10

- B) -10
 - C) 7
 - D) -7
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Q25. DIFFICULTY: HARD

If $f(x) = (2x + 1)/(x - 3)$, what is the value of $f(f(5))$?

- A) 5
 - B) $11/2$
 - C) $24/5$
 - D) Undefined
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SECTION B — CREATING LINEAR MODELS FROM WORD PROBLEMS (Q26–Q50)

Q26. DIFFICULTY: EASY

A taxi charges a flat fee of \$3.00 plus \$2.50 per mile driven. Which equation represents the total cost C , in dollars, for a trip of m miles?

- A) $C = 2.50m$
 - B) $C = 3.00m + 2.50$
 - C) $C = 2.50m + 3.00$
 - D) $C = 3.00 + 2.50 + m$
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Q27. DIFFICULTY: EASY

A candle is 18 centimeters tall when first lit. It burns at a constant rate and loses 2.4 centimeters every hour. Which equation represents the height h , in centimeters, of the candle after t hours?

- A) $h = 2.4t + 18$
 - B) $h = 18t - 2.4$
 - C) $h = 18 - 2.4t$
 - D) $h = 2.4 - 18t$
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Q28. DIFFICULTY: EASY

Marcus earns \$12.50 per hour working at a bookstore. He also received a one-time signing bonus of \$75 when he was hired. Which equation represents his total earnings E , in dollars, after working h hours?

- A) $E = 12.50h$
- B) $E = 75h + 12.50$
- C) $E = 12.50 + 75h$

D) $E = 12.50h + 75$

Q29. DIFFICULTY: EASY

A tank contains 240 gallons of water. Water drains from the tank at a constant rate of 15 gallons per minute. Which equation represents the volume V of water remaining in the tank after m minutes, where the tank is not yet empty?

- A) $V = 15m$
 - B) $V = 240 - 15m$
 - C) $V = 15 - 240m$
 - D) $V = 240m - 15$
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Q30. DIFFICULTY: EASY

A gym charges a \$40 enrollment fee and \$25 per month for membership. Which equation represents the total cost C , in dollars, of a membership lasting m months?

- A) $C = 65m$
 - B) $C = 40m + 25$
 - C) $C = 25m + 40$
 - D) $C = 40 + 25 + m$
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Q31. DIFFICULTY: MEDIUM

A plumber charges \$85 for the first hour of a service call and \$60 for each additional hour. Which equation represents the total charge T , in dollars, for a job that takes h hours, where $h \geq 1$?

- A) $T = 60h + 85$
 - B) $T = 60h + 25$
 - C) $T = 85h + 60$
 - D) $T = 145h$
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Q32. DIFFICULTY: MEDIUM

The temperature of a freezer is currently 8°F . A technician sets it to cool at a constant rate of 3°F every 10 minutes. Which equation represents the temperature T , in $^{\circ}\text{F}$, after x minutes, and after how many minutes will the temperature first reach -4°F ?

- A) $T = 8 - 0.3x$; 40 minutes
 - B) $T = 8 + 0.3x$; 40 minutes
 - C) $T = 8 - 3x$; 4 minutes
 - D) $T = 8 - 0.3x$; 30 minutes
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Q33. DIFFICULTY: MEDIUM

Two friends are hiking on separate trails. Alicia has already hiked 6 miles and continues at 3 miles per hour. Ben starts at the trailhead and hikes at 4.5 miles per hour. Which equation can be used to find the number of hours h after which Ben will have hiked the same total distance as Alicia?

- A) $3h + 6 = 4.5h$
 - B) $3h = 4.5h + 6$
 - C) $6h + 3 = 4.5h$
 - D) $4.5h + 6 = 3h$
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Q34. DIFFICULTY: MEDIUM

A cell phone plan charges \$0.05 per text message and a flat monthly fee of \$29.99. Which of the following represents the number of text messages t that can be sent in a month for a total bill of exactly \$34.99?

- A) $0.05t = 34.99$
 - B) $0.05t + 29.99 = 34.99$
 - C) $0.05 + 29.99t = 34.99$
 - D) $29.99t - 0.05 = 34.99$
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Q35. DIFFICULTY: MEDIUM

A printing company charges \$0.08 per page for the first 500 pages and \$0.05 per page for each additional page beyond 500. Which equation represents the total cost C , in dollars, of printing n pages where $n > 500$?

- A) $C = 0.08n$
 - B) $C = 0.05n + 15$
 - C) $C = 0.08n + 0.05(n - 500)$
 - D) $C = 0.08(n - 500) + 40$
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Q36. DIFFICULTY: MEDIUM

A researcher recorded the population P of bacteria in a dish, measured in thousands. She found that after t hours, $P = 3.2t + 5.6$. According to this model, what does the value 5.6 represent?

- A) The rate at which the bacteria population increases each hour
 - B) The total number of bacteria after one hour
 - C) The initial bacteria population, in thousands, when the experiment began
 - D) The number of hours it takes for the population to reach 5,600
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Q37. DIFFICULTY: MEDIUM

A car rental company models the total cost of renting a car as $C = 0.22d + 35$, where C is the total cost in dollars and d is the number of miles driven. What is the best interpretation of the number 0.22 in this context?

- A) The fixed daily rental fee, in dollars
 - B) The total cost after driving 1 mile
 - C) The cost per mile driven, in dollars
 - D) The number of miles included before additional charges apply
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Q38. DIFFICULTY: MEDIUM

An electrician earns \$95 per hour. She must purchase \$230 worth of materials for a job. If she wants her total earnings from the job to be at least \$1,000 after deducting the material costs, which inequality models the minimum number of hours h she must work?

- A) $95h \geq 1,000$
 - B) $95h - 230 \geq 1,000$
 - C) $95h + 230 \geq 1,000$
 - D) $95 - 230h \geq 1,000$
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Q39. DIFFICULTY: HARD

Company A rents storage units for \$120 per month with a one-time registration fee of \$50. Company B rents storage units for \$90 per month with a one-time registration fee of \$170. After how many months m will the total cost of renting from Company A equal the total cost from Company B?

- A) 2
 - B) 3
 - C) 4
 - D) 6
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Q40. DIFFICULTY: HARD

A factory produces widgets at a constant rate. At 9:00 AM, the inventory shows 840 widgets. By 1:00 PM, the inventory shows 1,160 widgets. Assuming production is constant and no widgets are removed, which function $W(t)$ represents the number of widgets t hours after 9:00 AM?

- A) $W(t) = 80t + 840$
 - B) $W(t) = 80t + 760$
 - C) $W(t) = 320t + 840$
 - D) $W(t) = 80t + 1,160$
-

Q41. DIFFICULTY: HARD

The linear function f models the value V , in dollars, of a piece of equipment over time. The equipment was worth \$8,400 when purchased and loses value at a constant annual rate. After 6 years it is worth \$5,100. What is the best interpretation of the slope of f in this context?

- A) The equipment loses \$550 in value each year
- B) The equipment loses \$5,100 in value each year

- C) The equipment was purchased for \$8,400
 - D) The equipment will be worth \$0 after approximately 15.3 years
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Q42. DIFFICULTY: HARD

A school cafeteria sells lunches for \$3.75 each. The cafeteria's daily operating costs are \$210. The cafeteria must sell at least how many lunches per day to cover its daily operating costs?

- A) 54
 - B) 55
 - C) 56
 - D) 57
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Q43. DIFFICULTY: HARD

A scientist is tracking the decline in a whale population. In 2010 there were 1,240 whales. The population decreased by 48 whales per year. In what year will the model predict the whale population to first fall below 800?

- A) 2018
 - B) 2019
 - C) 2020
 - D) 2021
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Q44. DIFFICULTY: HARD

A moving company charges a base fee of \$ b and a rate of \$ r per mile. If the charge for a 20-mile move is \$150 and the charge for a 35-mile move is \$195, what is the base fee b ?

- A) \$30
 - B) \$60
 - C) \$90
 - D) \$120
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Q45. DIFFICULTY: HARD

Two water tanks are being filled simultaneously. Tank A contains 120 liters and is being filled at 8 liters per minute. Tank B contains 105 liters and is being filled at 11 liters per minute. After how many minutes will Tank B contain exactly 15 liters more than Tank A?

- A) 5 minutes
 - B) 10 minutes
 - C) 15 minutes
 - D) 20 minutes
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Q46. DIFFICULTY: MEDIUM

The number of students S enrolled in an after-school program is modeled by $S = -3w + 74$, where w is the number of weeks since the program began. What is the best interpretation of the value 74 in this context?

- A) The number of students who leave the program each week
 - B) The total number of students enrolled after 74 weeks
 - C) The number of students enrolled when the program began
 - D) The week in which enrollment first reaches zero
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Q47. DIFFICULTY: MEDIUM

A delivery driver is paid \$0.40 per mile driven plus a daily guarantee of \$80. On a day when the driver is paid \$180, how many miles did the driver drive?

- A) 200 miles
 - B) 250 miles
 - C) 300 miles
 - D) 450 miles
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Q48. DIFFICULTY: MEDIUM

A pool is being drained. After 2 hours, the pool contains 4,400 gallons. After 5 hours, the pool contains 2,900 gallons. Assuming the pool drains at a constant rate, which equation represents the gallons G remaining after t hours?

- A) $G = -500t + 5,400$
 - B) $G = -500t + 4,400$
 - C) $G = 500t + 5,400$
 - D) $G = -1,500t + 7,400$
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Q49. DIFFICULTY: HARD

A salesperson earns a weekly base salary plus a commission for each unit sold. When she sells 10 units in a week, she earns \$860. When she sells 25 units, she earns \$1,160. What would she earn in a week when she sells 40 units?

- A) \$1,360
 - B) \$1,460
 - C) \$1,560
 - D) \$1,760
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Q50. DIFFICULTY: HARD

The linear model $P = 1.6t + 42$ estimates the number of pounds P of recycled material collected by a community t weeks after a recycling drive began. According to the model, how many more pounds of recycled material were collected in week 8 than in week 3?

- A) 4 pounds
 - B) 8 pounds
 - C) 16 pounds
 - D) 5 pounds
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— END OF QUESTIONS —